7 DIGITS display

IC14 74HC373
IC15 74HC373
IC16 74HC373
IC17 74HC373
LED2 GL7Q220.6
LED3 GL7Q220.6
IO_D0
IO_D1
IO_D2
IO_D3
IO_D4
IO_D5
IO_D6
IO_D7
DIGIT0
DIGIT1
DIGIT2
DIGIT3
RESET

RA18 10K
RA19 10K
IC29 74AHC244
IC30 74AHC244
DIPSW1
DIPSW2
IC18A 74AHC04
IC18B 74AHC04

IC11 74HC245
IC5A 74LS139
IC39 74HC238
IC13 74HC238
IC12 74HC238
RA9 10K
IC35.11
IC36.11
IC37.11
IC38.11
Not populated IC
OPTO1_IN
OUTG1
OUTG3
OPTO2_IN
OUTG2
OUTG4
IORQ
WR
RD
IC2B 74HC00

IC33 M6253
CLK4
IC4 GAL16V8
ROM
RAM
IC12 25 (display, 70pin)
ADC (wheel position, pedal ?)
IC33.4 (power input/output)
IC39.4 (nothing here)
J1
code 315-3060

IC19 TD62384
IC20 TD62384
IC21 TD62384
IC22 TD62384
IC23 74HC373
IC24 74HC373
IC25 74HC373
IC26 74HC373
CN2
CN4
CN7

C100 C0
C99 470µ/10
C101 1µ
EMF16
EMF19
D2 4148
D3 4148
D4 4148
D5 4148
D6 4148
D7 4148
D8 4148
D9 4148
R21 820
R22 820
R23 820
R24 820
C96 1n
C95 1n
C94 1n
C93 1n
IC34A CD4066BE
IC34D CD4066BE
IC34B CD4066BE
IC34C CD4066BE
IC18F 74AHC04
IC18E 74AHC04
PA6

IC10 74HC245
DATA BUS INTERNAL
DATA BUS EXTERNAL
TMP284C015BF IC3PORTA
TMP284C015BF IC3PORTB
CTCA
CTCB
PORTA
PORTB
RA2 10K
RA3 10K
RA6 10K
SYNCA
RXDA
TXDA
DTRA
RTSA
CTSA
DCDA
SYNCB
RXDB
TXDB
DTRB
RTSB
CTSB
DCDB
ATSB
BSTB
W/RDYA
W/RDYB
ARDY
BRDY
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7

IC1 4040
OSC1 C-1H 16.000
CLK0
CLK4
Reset
EMF9
EMF10
EMF11
EMF12
L1
L2

IC3CLOCK_PWR
IC3CPU
IC3CTC
CLK/TRIG0 ZC/TO0
CLK/TRIG1 ZC/TO1
CLK/TRIG2 ZC/TO2
CLK/TRIG3 ZC/TO3
XTAL1
XTAL2
CLKIN
CLKOUT
RESET
NMI
IEO
IEI
M1
MREQ
IORQ
RD
WR
RFSH
HALT
WAIT
INT
BUSRQ
BUSACK
WDTOUT
EV
RA5 10K
RA4 10K
IEI
BSTB
NMI
WDTOUT
EX
MREQ
WR
IORQ
ATSB
INT
HALT
BUSACK
BUSRQ
RFSH
JPSW2 Jumper_3_Open
JPSW1 Jumper_3_Open
JPSW3 Jumper_3_Open
Deactivate watchdog
L3
L4
L5
L6
EMF5
EMF6
EMF7
EMF8
EMF13
EMF14
EMF15
EMF17
EMF18
EMF20
EMF21
EMF22
EMF23

WDTOUT
RESET
IC7 MB3771
IC2D 74HC00
IC2A 74HC00
C46 1µ
R1 2K7
R2 1K
C70 1µ
C47 470µ/16v
FL1 BNX02
R3 470
D10 LED
C48 470µ/16v
CN8

RA20 10K
RA21 10K
R5 270
R6 270
R7 270
R8 270
R9 270
R10 270
R11 270
R12 270
R13 270
R14 270
R15 270
R16 270
R17 270
R18 270
R19 270
R20 270
PC8A
PC7B
PC6C
PC5D
PC4A
PC3B
PC2C
PC1D
PC16A
PC15B
PC14C
PC13D
PC12A
PC11B
PC10C
PC9D
TLP521
CN1
CN3
IC27 74AHC244
IC28 74AHC244
IC8B 74AHC04
IC8D 74AHC04

D1 4004
basic RAM saving with supercap and isolation while in reset
IC9 TC55257DFI-85
IC8E 74LS126
IC8F 74LS126
IC8C 74LS126
IC8D 74LS126
IC8A 74LS126
IC8 27C512
IC8 74LS139
RA1 10K
C52 C0
CM1 SuperCap
D0
D1
D2
D3
D4
D5
D6
D7
RESET
RAM
ROM

IC2E 74HC00
IC18G 74AHC04
IC5C 74LS139
IC34E CD4068EE
RA7
CN6
CN9
IC35 74HC373
IC36 74HC373
IC37 74HC373
IC38 74HC373

Decoupling caps are not listed
Some unpopulated components are present on the schematic
EMI are not all in place
Pins without NC symbol (X) need lift-up of the chip to see tracks or really NC.
Feedback from donor board welcomed

https://github.com/Franck78/SEGA_838-12898
https://buymeacoffee.com/franck78

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File: sega-838-12898.kicad_sch

Title: SEGA Model 3 I/O Driver Board #838-12898

Size: A1 | Date: 2026-11-08 | Rev: v1.0
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